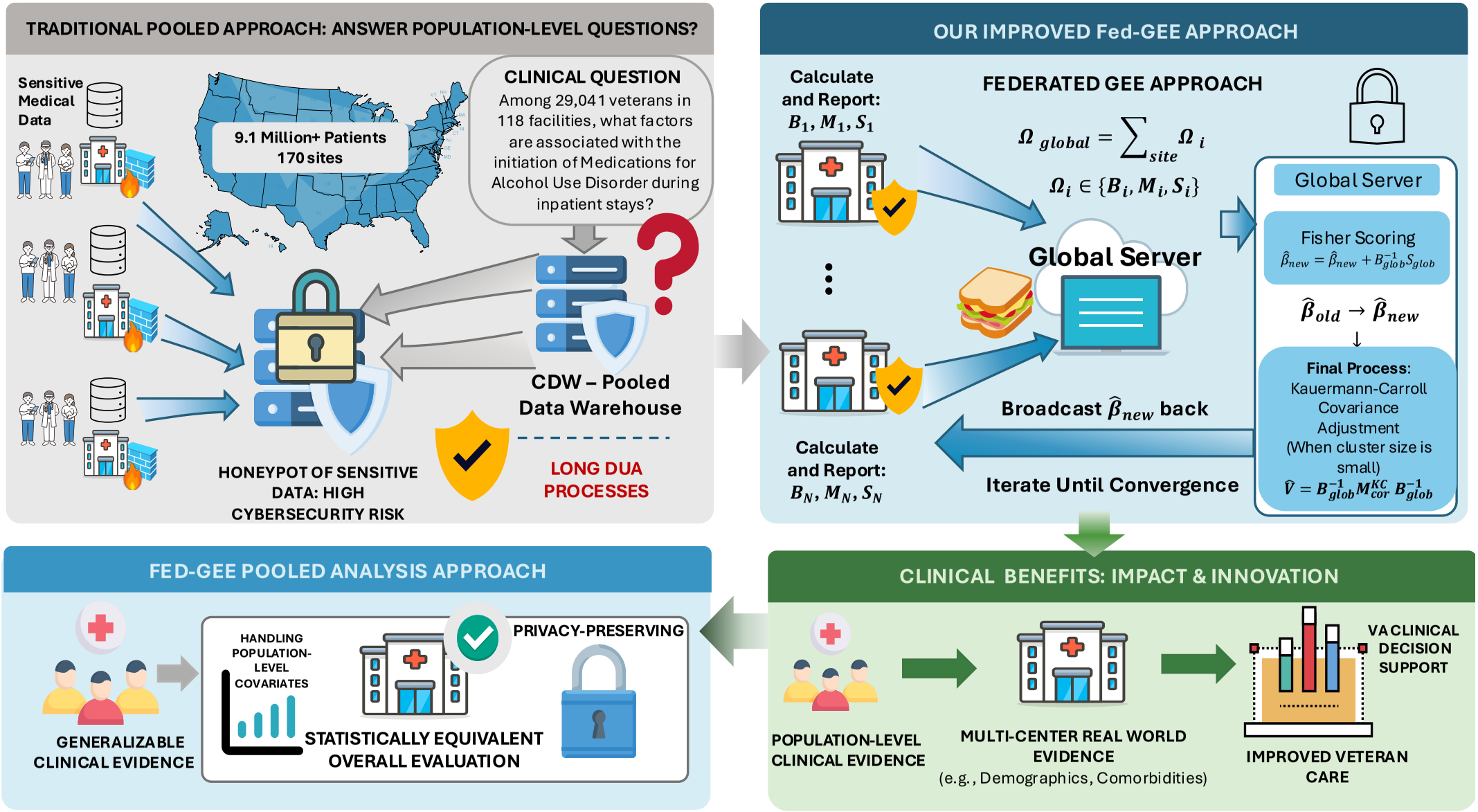
TRADITIONAL POOLED APPROACH: ANSWER POPULATION-LEVEL QUESTIONS?
Sensitive Medical Data
9.1 Million+ Patients
170 sites
CLINICAL QUESTION
Among 29,041 veterans in 118 facilities, what factors are associated with the initiation of Medications for Alcohol Use Disorder during inpatient stays?
CDW – Pooled Data Warehouse
HONEYPOT OF SENSITIVE DATA: HIGH CYBERSECURITY RISK
LONG DUA PROCESSES
OUR IMPROVED Fed-GEE APPROACH
Calculate and Report: B_1, M_1, S_1
FEDERATED GEE APPROACH
$\Omega_{global} = \sum_{site} \Omega_i$
$\Omega_i \in \{B_i, M_i, S_i\}$
Global Server
Fisher Scoring
$\hat{\beta}_{new} = \hat{\beta}_{new} + B_{glob}^{-1} S_{glob}$
$\hat{\beta}_{old} \rightarrow \hat{\beta}_{new}$
Final Process: Kauermann-Carroll Covariance Adjustment (When cluster size is small)
$\hat{V} = B_{glob}^{-1} M_{cor}^{KC} B_{glob}^{-1}$
Broadcast $\hat{\beta}_{new}$ back
Calculate and Report: B_N, M_N, S_N
Iterate Until Convergence
FED-GEE POOLED ANALYSIS APPROACH
GENERALIZABLE CLINICAL EVIDENCE
HANDLING POPULATION-LEVEL COVARIATES
PRIVACY-PRESERVING
STATISTICALLY EQUIVALENT OVERALL EVALUATION
CLINICAL BENEFITS: IMPACT & INNOVATION
POPULATION-LEVEL CLINICAL EVIDENCE
MULTI-CENTER REAL WORLD EVIDENCE
(e.g., Demographics, Comorbidities)
VA CLINICAL DECISION SUPPORT
IMPROVED VETERAN CARE